

What the Equations Are Saying: A Notation Companion to Proof-Guided Branch-Aware Search

BSD (bsd.developer@proton.me)\ Independent Researcher

30 September 2026

Abstract

This companion explains the mathematical notation in *Proof-Guided Branch-Aware Search for a New Matrix Multiplication Bound*. It follows that paper's equations in order. For each one, it identifies the objects, operations, and logical status of the statement, then develops a simpler picture of why the expression matters. Additional equations here are explanatory, not new claims about the measured search or a substitute for its exact certificate.

1. How to read this paper

The methods paper alternates between four kinds of object. A **physical point** θ is an assignment of distributions and scalar parameters. A **score** such as $E(\theta)$ or $F_{\text{hard}}(\theta; \Omega)$ is a number computed from that assignment. A **proposition** such as $H_i(\xi)$ is something that could be true or false. A **proof** such as π_i is evidence that a particular proposition is true. Keeping those types apart prevents the most tempting misreading: a good numerical score is not itself a proof of a bound.

Here is the notation used repeatedly below.

Mark	Mathematical reading
$g(\theta)$	The value of the function g at the parameter assignment θ .
$g_i, v_{i,k}$	Subscripts label an object or its position; i is a site and k is a branch.
θ_0, θ_1	An origin and a proposed endpoint. A subscript 0 does not mean zero.
θ_*, ξ_*	A selected, particular witness. A star is a label, not multiplication.
Δg	The finite change $g(\theta_1) - g(\theta_0)$, not a derivative.
$\sum, \min, \max$	Add a finite collection; take its smallest value; take its largest value.
$\forall, \wedge, \implies$	“For every”; “all of these hold”; “if the left side holds, then the right side holds.”
$\ v\ ^2$	The squared Euclidean length of a vector v , when the chosen local coordinates are Euclidean.
$\underline{x}, \bar{x}$	A certified lower and upper enclosure of a real quantity x , respectively.

The word **hard** refers to the original objective with actual minima over branches. **Mix** refers to a smoother weighted-mixture expression. Their values agree only when the relevant mixture slack is zero. The word **exact** refers to rational input and directed checking; it does not describe a floating-point search readout.

2. The finite object behind the exponent

The matrix multiplication exponent ω is the unknown quantity to be bounded. The paper fixes the Coppersmith–Winograd base parameter $q = 5$ and the maximum recursion level $\ell^* = 4$ (Coppersmith and Winograd 1990; Alman et al. 2025). The star on ℓ^* means the chosen top level. At an intermediate level ℓ , a shape is a triple of nonnegative integers,

$$(a_X, a_Y, a_Z), \quad a_X + a_Y + a_Z = 2^\ell.$$

Term	Precise meaning
a_X, a_Y, a_Z	The three coordinates of a shape; each belongs to $\mathbb{N}_0$.
2^ℓ	The prescribed coordinate total at level ℓ .
$\ell^* = 4$	The largest recursion level in this instance; it does not say that every tree node has level four.

The tree records successive choices of regions, shapes, splits, and leaf parameters. Its paths matter: two nodes with the same shape and level can still represent different choices. The symbol θ collects every free distribution and leaf parameter into one large assignment. It is a point in a constrained parameter space Θ , not merely a short vector in the displayed equations.

One can picture a shape as an address with three entries whose total is fixed by the floor of the tree. The parameter θ is the full set of instructions placed at all those addresses. The rest of the paper asks whether a particular complete set of instructions passes a stringent inequality.

2.1 The fixed threshold

The methods paper abbreviates a constant:

$$C_{q,\ell^*} = 2^{\ell^*-1} \log_2(q+2) = 8 \log_2 7.$$

Term	Precise meaning
C_{q,ℓ^*}	A threshold fixed once q and ℓ^* are fixed. Its subscripts record that dependence.
2^{ℓ^*-1}	A power of two arising from the chosen recursion level; at $\ell^* = 4$ it equals $2^3 = 8$.
$\log_2(q+2)$	Base-two logarithm of $q+2$; with $q = 5$, its argument is 7.
$=$	The second expression is an exact substitution, not a numerical approximation.

The base-two logarithm asks how many doublings are needed to reach a number. This constant sets the height of the bar. Search changes the assignment θ ; it does not move the bar $8 \log_2 7$.

3. The feasibility inequality and the hard score

The assignment θ determines two real quantities: a retained exponent $E(\theta)$ and a matrix-size exponent $M(\theta)$. A proposed bound $\Omega \geq 0$ must satisfy equation (1) of the methods paper:

$$E(\theta) + \Omega M(\theta) \geq C_{q,\ell^*}. \tag{1}$$

Term	Precise meaning
$E(\theta)$	The retained-exponent contribution of this complete assignment. Its internal formula includes branch minima.
$M(\theta)$	The matrix-size contribution of the same assignment; it also has a limiting three-way minimum.
Ω	A candidate numerical <i>upper bound</i> to be justified, distinct from the unknown exponent ω .
$\Omega M(\theta)$	Ordinary multiplication. The candidate bound is weighted by the matrix-size term.
$\geq C_{q,\ell^*}$	The feasibility test: the left side must reach the fixed threshold.

Equation (1) is not the definition of ω . It is a condition in the combination-loss formulation. The cited mathematical theorem says that a fully feasible assignment with candidate Ω implies $\omega \leq \Omega$ (Alman et al. 2025). That implication is a separate logical step.

For intuition, call E the contribution already earned and M the leverage on the rate Ω . The inequality asks whether the earned part plus the rate-dependent part reaches the bar. This picture is useful, but the exact definitions of E and M remain those of the combination-loss problem.

3.1 Solving for a candidate rate

If $M(\theta) > 0$, ordinary algebra turns (1) into the smallest nonnegative candidate at this particular point:

$$\Omega_{\text{hard}}(\theta) = \max \left\{ 0, \frac{C_{q,\ell^*} - E(\theta)}{M(\theta)} \right\}. \quad (2)$$

Term	Precise meaning
$\Omega_{\text{hard}}(\theta)$	The hard numerical score of the assignment. The subscript says actual branch minima are used.
$C_{q,\ell^*} - E(\theta)$ $/M(\theta)$	The remaining shortfall after the retained contribution. Division by a positive matrix-size term; this is why the assumption $M(\theta) > 0$ matters.
$\max\{0, \cdot\}$	Enforces the stated restriction $\Omega \geq 0$. A negative shortfall gives score zero, not a negative candidate bound.

To derive it, subtract $E(\theta)$ from both sides of (1), then divide by $M(\theta) > 0$ without reversing the inequality. If $E(\theta)$ has already reached the bar, the formula's zero is only a property of this feasibility score; it is not a claim that matrix multiplication has exponent zero.

Think of (2) as a price per unit of leverage: take the gap still to be closed and divide it by how strongly Ω acts. Search can reduce the price by closing the gap, increasing the useful leverage, or both. It must evaluate the *hard* expression to know whether either happened.

4. Why a smooth mixture can lie

At a branch site i , let $v_{i,1}(\theta)$, $v_{i,2}(\theta)$, and $v_{i,3}(\theta)$ be the three competing real values. Choose weights $w_{i,k}(\theta) \geq 0$ with $\sum_{k=1}^3 w_{i,k}(\theta) = 1$. Equation (3) defines the weighted mixture b_i and its slack s_i :

$$b_i(\theta) = \sum_{k=1}^3 w_{i,k}(\theta) v_{i,k}(\theta), \quad s_i(\theta) = b_i(\theta) - \min_k v_{i,k}(\theta) \geq 0. \quad (3)$$

Term	Precise meaning
i	A site at which three branch contributions compete.
$k \in \{1, 2, 3\}$	The branch index at that site.
$v_{i,k}(\theta)$	The value of branch k at site i for this assignment.
$w_{i,k}(\theta)$	A nonnegative mixture weight. The three weights sum to one and may themselves change with θ .
$b_i(\theta)$	A convex weighted average of the branch values.
$\min_k v_{i,k}(\theta)$	The branch value the hard minimum actually retains.
$s_i(\theta)$	The excess of the mixture over the minimum; it is never negative under the stated weight conditions.

Why is $s_i \geq 0$? Every $v_{i,k}$ is at least the minimum m_i . Multiplying these three inequalities by nonnegative weights and adding gives $b_i \geq m_i \sum_k w_{i,k} = m_i$. This is an exact algebraic fact, not a local approximation.

For a toy example, take branch values (10, 9, 8) and equal weights. The mixture says 9, while the hard minimum says 8, so the slack is 1. The average is not *wrong*; it answers a different question. The hard problem pays attention to the weakest branch. Slack is the exact price of looking at the average instead.

4.1 The ledger identity

With all site coefficients and other terms accounted for, the methods paper writes the hard surplus at a fixed candidate Ω_0 as

$$F_{\text{hard}}(\theta; \Omega_0) = F_{\text{mix}}(\theta; \Omega_0) - S(\theta; \Omega_0). \quad (4)$$

Term	Precise meaning
$F_{\text{hard}}(\theta; \Omega_0)$	The surplus for the original minimum-based objective at fixed Ω_0 . Nonnegative surplus means its inequality is met, subject to domain conditions.
$F_{\text{mix}}(\theta; \Omega_0)$	The corresponding expression using the selected convex mixtures.
$S(\theta; \Omega_0)$	Aggregate branch slack, including the relevant nonnegative coefficients. It is not just one s_i .
Ω_0	A fixed candidate rate while comparing points; the subscript 0 here does not make it a branch index.

The identity is *pointwise*: it holds at each represented assignment. The weights, branch values, coefficients, and root masses can all change when θ changes. Subtracting the identity at θ_0 from the one at θ_1 gives equation (5):

$$\Delta F_{\text{hard}} = \Delta F_{\text{mix}} - \Delta S. \quad (5)$$

Term	Precise meaning
ΔF_{hard}	$F_{\text{hard}}(\theta_1; \Omega_0) - F_{\text{hard}}(\theta_0; \Omega_0)$.
ΔF_{mix}	The analogous finite change in mixture surplus.
ΔS	The endpoint slack minus origin slack, with each point's own weights and coefficients.
—	Exact subtraction of changes; no Taylor remainder is hidden in (5).

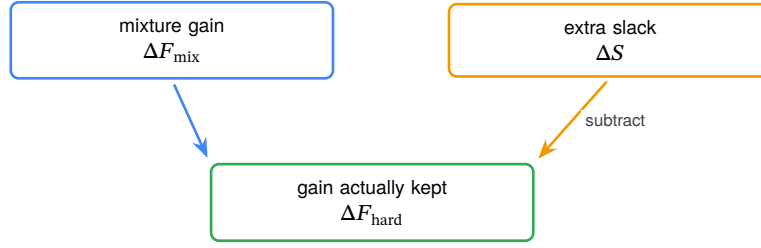


Figure 1: The branch ledger. The smooth model’s apparent gain becomes hard gain only after the change in branch slack is charged.

If the mixture gains 2 units while slack grows 3 units, the hard surplus falls by 1 unit. If slack shrinks, the hard result can improve even when the mixture barely moves. This is why a correction that drives branches toward equality can matter more than another nominally good gradient step. The paper’s measurements are finite and numerical; the identity explaining them is exact.

5. Geometry near one point

The paper studies local directions at a validated origin θ_0 . Three branches have two independent differences: for example, $c_{i,1} = v_{i,1} - v_{i,3}$ and $c_{i,2} = v_{i,2} - v_{i,3}$. These are **contrasts**. If both vanish, all three branch values agree. Twenty relevant three-way sites give forty such contrasts. Collect them as a vector $c(\theta)$, and write J for its Jacobian at θ_0 :

$$J = \left. \frac{\partial c}{\partial \theta} \right|_{\theta_0}, \quad \ker J = \{d : Jd = 0\}.$$

Term	Precise meaning
$c(\theta)$	The vector of branch differences at all tracked sites.
J	The linear map of first derivatives of those differences at the stated origin. Its origin is part of its identity.
d	A proposed displacement in the physical parameter coordinates.
Jd	Predicted <i>first-order</i> change of the contrasts under displacement d .
$\ker J$	The null space of J : directions with zero predicted first-order contrast change.

A direction in $\ker J$ is tangent to the local balance conditions. It does not promise exact balance after a finite step: curvature can bend the path away. If $\tilde{F}$ denotes the differentiable local objective, and $P_{\ker J}$ the projection into that tangent space, the local balanced gradient can be pictured as

$$g_{\text{bal}} = P_{\ker J} \nabla \tilde{F}(\theta_0).$$

Term	Precise meaning
$\nabla \tilde{F}(\theta_0)$	Vector of first derivatives of the local smooth objective at the actual origin.
$P_{\ker J}$	A projection onto the rank-checked balance tangent space.
g_{bal}	The component of the gradient available without changing tracked contrasts to first order.

The projected gradient is an arrow on a map of the local terrain. The tangent condition says the arrow initially travels along the ridge where competing branches stay level. A short arrow is only a local prediction; the finite endpoint must still be tested against the real minimum. The paper’s reported 4.594% captured squared norm means an existing twelve-direction basis represented little of that fresh balanced arrow’s energy. It motivated more directions, not a certified prediction of a large gain.

5.1 Curvature without a full Hessian

The paper estimates curvature using first gradients at displaced points. For exposition, put $G(\theta) = \nabla \tilde{F}(\theta)$. A central difference in direction d has the form

$$Hd \approx \frac{G(\theta_0 + hd) - G(\theta_0 - hd)}{2h}, \quad h > 0.$$

Term	Precise meaning
H	The Hessian of the smooth local objective, where that Hessian exists; the method need not materialize its full matrix.
Hd	A Hessian-vector product, the change in gradient predicted per unit displacement along d .
h	A finite probe distance, not the proof-obligation symbol H_i .
$G(\theta_0 \pm hd)$	Fresh first gradients at two actual nearby physical points.
$\approx$	A finite-difference estimate, not an equality or a certified interval bound.

The scalar $d^T Hd$ describes curvature along one direction. Using two probe distances checks whether the estimate is stable under a change of scale. It cannot prove that curvature stays bounded throughout a finite move. In ordinary language, the method tests how the slope changes a little ahead and a little behind, instead of building an enormous map of every possible pair of directions. The paper also reports why this mattered: a native second-derivative route failed a zero-curvature diagnostic, so it was not accepted as the local oracle.

5.2 A finite step is a different question

A local model proposes a step ηd and perhaps a normal correction δ to reduce branch contrasts. At the linearized level, a zero-contrast correction would try to solve

$$J\delta \approx -c(\theta_0 + \eta d).$$

Term	Precise meaning
η	A finite step size or fraction of a proposed displacement.
$\theta_0 + \eta d$	The trial physical point before correction.
δ	An additional displacement chosen to reduce contrast.
$-c(\theta_0 + \eta d)$	The change that would cancel the observed contrasts in a linear approximation.
$\approx$	The Jacobian equation describes a local correction model, not exact equality at the corrected endpoint.

The paper also describes corrections that restore inherited contrasts rather than drive them to zero. The target matters: preserving an old imbalance can preserve an avoidable slack bill. A correction is accepted only after checking the full finite point, its domain constraints, and its hard score. Geometry suggests where to try; the endpoint decides.

6. Proof obligations: turning one huge claim into smaller questions

Fix a rational target τ with $0 \leq \tau < 2.371177$. At this fixed target, the exact surplus can be written

$$F_\tau(\theta) = E(\theta) + \tau M(\theta) - C_{q,\ell^*}.$$

Term	Precise meaning
τ	The target bound to be established; it is fixed while the obligation is analyzed.
$F_\tau(\theta)$	Left side minus right side of equation (1) when $\Omega = \tau$.
$F_\tau(\theta) \geq 0$	The numeric inequality needed for feasibility, in addition to all domain constraints.

This is a useful change of viewpoint. Instead of asking directly for a small exponent, we ask whether the exact assignment leaves a nonnegative surplus at a chosen target. If the surplus is negative, the target fails at that point, no matter how attractive a surrogate score looks.

The paper collects local propositions as $\mathcal{H} = (H_1, \dots, H_k)$. An extended witness $\xi = (\theta, d, \eta, w, \dots)$ may contain a physical point, directions, step sizes, auxiliary weights, and other evidence. The notation $\theta(\xi)$ means “take the physical assignment out of that larger package.” Equation (6) is a *conditional* theorem:

$$\Pi_{\tau, \mathcal{H}} : \quad \forall \xi, \quad \left(\bigwedge_{i=1}^k H_i(\xi) \right) \implies F_\tau(\theta(\xi)) \geq 0. \quad (6)$$

Term	Precise meaning
$\mathcal{H} = (H_1, \dots, H_k)$	A finite list of propositions chosen to make the target inequality derivable.
$H_i(\xi)$	Obligation i evaluated at extended witness ξ ; for example, a domain, curvature, correction, or slack premise.
$\Pi_{\tau, \mathcal{H}}$	A proof of the general implication, with the target and obligation family fixed.
$\forall \xi$	The implication holds for every extended witness satisfying its premises, not merely for a sampled endpoint.
$\bigwedge_{i=1}^k H_i(\xi)$	Every obligation must be true. One failed leaf blocks this route.
$\implies$	Logical implication. It does not assert that the premises have already been proved.
$\theta(\xi)$	Projection from the extended witness to the actual physical parameters.

This is the place where the research method becomes unusual. A theorem like (6) does not yet lower ω . It describes what evidence *would* make the target work. The obligations are like missing pieces in a machine: Lean can prove that, if every piece fits, the machine runs; search and further proofs must still supply the pieces. They are “variables” in research design because their decomposition and thresholds may be revised, but they are logical claims, not numerical coordinates that gradient descent can tune.

The missing pieces are genuinely useful only if they are easier to check than the original whole claim and reveal why a candidate fails. Taking $H_1(\xi)$ to mean “the final theorem is true” would make (6) easy to write and useless to use.

6.1 Supplying the pieces

For a selected exact extended witness ξ_* , equation (7) shows what must be present before the external feasibility-to-exponent theorem can be applied:

$$\begin{aligned} & \pi_{\text{dom}} : \text{Dom}(\theta_*), & \pi_i : H_i(\xi_*) \quad (1 \leq i \leq k), \\ (\Pi_{\tau, \mathcal{H}}, \pi_{\text{dom}}, \pi_1, \dots, \pi_k) \implies \text{Feasible}(\theta_*, \tau) \implies \omega \leq \tau. \end{aligned} \quad (7)$$

Term	Precise meaning
ξ_*, θ_*	The selected exact extended witness and its physical assignment.
$\text{Dom}(\theta_*)$	The conjunction of all required domain constraints on the physical assignment.
π_{dom}	A proof of those domain constraints.
$\pi_i : H_i(\xi_*)$	A proof term whose type is obligation i at the selected witness. A sampled numerical value is not such a term.
$\text{Feasible}(\theta_*, \tau)$	Domain conditions together with nonnegative target surplus.
$\omega \leq \tau$	The desired bound, reached using the separately cited combination-loss implication.

Read (7) from left to right as a proof assembly line. The structural theorem says how the pieces combine. Each π_i supplies one piece. The domain proof ensures the physical object is legal. Only after the assembled feasibility statement is obtained does the cited mathematical theorem turn it into a statement about ω . No assumption may be smuggled across that last arrow.

7. How much improvement is actually needed?

At an origin θ_0 , put $M_0 = M(\theta_0) > 0$. The normalized shortfall at target τ is equation (8):

$$D_\tau = \frac{8 \log_2 7 - E(\theta_0) - \tau M(\theta_0)}{M_0}. \quad (8)$$

Term	Precise meaning
D_τ	The origin's target-surplus deficit, divided by its positive matrix-size term.
$8 \log_2 7$	The same fixed threshold $C_{5,4}$ from Section 2.1.
$E(\theta_0) + \tau M(\theta_0)$	The left side of (1) evaluated at the origin and at target τ .
M_0	Shorthand for $M(\theta_0)$; positivity makes the division and inequality direction valid.

The identity beneath the notation is

$$F_\tau(\theta_0) = -M_0 D_\tau.$$

If $D_\tau > 0$, the origin is below the target bar and needs a gain. If $D_\tau \leq 0$, the inequality is already met at that origin, provided its domains are valid. Dividing by M_0 converts the deficit into the same rate-like units as a change in an exponent bound. The number is not a magic threshold: it is exactly the distance from the bar, measured using the origin's leverage.

7.1 A conditional gain bound

Equation (9) is the paper’s schematic finite-gain lemma. It applies only under its stated alignment, curvature, correction, and branch conditions:

$$\underbrace{\frac{\Delta F_{\text{hard}}}{M_0}}_{\text{needed hard gain}} \geq \underbrace{\eta(1 - K\eta/2)\|v\|^2}_{\text{modeled mixture gain}} - \underbrace{B_{\text{corr}}}_{\text{finite correction cost}} - \underbrace{B_{\text{slack}}}_{\text{branch imbalance cost}}. \quad (9)$$

Term	Precise meaning
$\Delta F_{\text{hard}}/M_0$	Actual finite change in hard surplus, normalized by the origin’s positive M_0 . For target closure, evaluate the surplus at the fixed target τ .
η	A finite step length.
v	A chosen aligned direction in the local model; $\ v\ ^2$ measures its squared size.
K	A curvature bound required by the conditional theorem, not merely a sampled Hessian estimate.
$\eta(1 - K\eta/2)\ v\ ^2$	A lower estimate of useful mixture gain under the lemma’s hypotheses.
B_{corr}	An allowance for the loss caused by making a finite branch correction.
B_{slack}	An allowance for adverse branch-slack change. The actual slack may instead shrink.
$\overset{\cdot}{\text{label}}$	A visual annotation naming a term; it adds no mathematical operation.

The factor $1 - K\eta/2$ is the familiar second-order caution: the first part of a step may help, but curvature can erode its linear promise. The two B terms charge what happens when we correct the finite move and when the hard minimum disagrees with the mixture. Every term is expressed in the same normalized units so it can be compared with D_τ .

Here is the exact logic of the comparison. If a proved lower bound on the right side of (9) reaches D_τ , then

$$\frac{\Delta F_{\text{hard}}}{M_0} \geq D_\tau \implies F_\tau(\theta_1) = F_\tau(\theta_0) + \Delta F_{\text{hard}} \geq 0.$$

The endpoint must still satisfy its domain conditions. Also, the paper’s sampled first gradients and finite differences suggest plausible values for K and the cost allowances; they do **not** prove the uniform real bounds needed to instantiate (9) for the winning floating trajectory.

The picture is now simple. The origin has a debt D_τ . A proposed move promises a gross gain. Curvature, correction, and branch imbalance send bills. Only the gain left after those bills can pay the debt. The identity (5) tells us why the branch bill is not optional.

8. The last step is a check, not a search story

The paper’s final result is attached to a rationalized, frozen witness, not to the optimizer path. Let $\hat{\theta}$ denote such an exact assignment. Directed arithmetic aims to prove a safe lower bound on the target surplus. Schematically, when the component signs permit the displayed directions,

$$\underline{E}(\hat{\theta}) + \tau \underline{M}(\hat{\theta}) - \overline{C} \geq 0 \implies F_\tau(\hat{\theta}) \geq 0 \implies \omega \leq \tau.$$

Term	Precise meaning
$\hat{\theta}$	A particular exact assignment, usually with rational parameters, derived from but not numerically identical to a floating point.
$\underline{E}, \underline{M}$	Certified lower enclosures of contributions used with positive coefficients. The full checker handles each signed subterm in its required direction.
$\overline{C}$	A certified upper enclosure of the threshold, including its logarithm.
≥ 0	A sufficient, safely rounded exact acceptance inequality.
$\omega \leq \tau$	The mathematical conclusion after domain checks and the cited feasibility theorem are applied.

The overline and underline are arrows of caution. If one wants to prove “left side at least right side,” one makes the left side pessimistically small and the right side pessimistically large. If even that conservative comparison passes, rounding cannot be the reason it passed. This is the opposite of asking a floating-point display whether two nearby decimals look ordered.

There are therefore three numbers one must not conflate: the smooth model’s prediction, the independently checked *numerical* hard score, and the exact rational bound named by the certificate theorem. The first guides a move. The second selects a candidate. The third, together with the stated trust assumptions, supports the claim about ω . This companion deliberately leaves that rational unnamed: it explains the notation, rather than replacing the result-local certificate and theorem.

9. One view of the whole argument

The story can be compressed into four questions, each stricter than the last. Does a point look promising under a local mixture? Does the finite point improve the minimum-based score after slack is charged? Can every needed statement about an exact witness be checked? Does the resulting feasibility theorem imply the displayed ω bound under its declared assumptions?

The equations answer these in order. Equations (1)–(2) define what a candidate must accomplish. Equations (3)–(5) explain why a smooth map can mislead. The Jacobian and curvature notation say where to look next, without pretending that a tangent or a finite difference is a finite proof. Equations (6)–(9) turn an apparent improvement into explicit obligations and a quantitative shortfall. The exact checker then tests one complete frozen assignment. The profound point is also the plain one: an optimizer may be clever about where to look, but the theorem must still be honest about what it has actually seen.

Bibliography

- Alman, Josh, Ran Duan, Virginia Vassilevska Williams, Yinzhan Xu, Zixuan Xu, and Renfei Zhou. 2025. “More Asymmetry Yields Faster Matrix Multiplication.” In *ACM-SIAM Symposium on Discrete Algorithms (SODA)*.
- Coppersmith, Don, and Shmuel Winograd. 1990. “Matrix Multiplication via Arithmetic Progressions.” *Journal of Symbolic Computation* 9 (3): 251–80.